Problem Statement : 01
SQL Queries
Consider the given Database Schema:

officers (officer-name, street, city)
placed (officer-name, company-name, salary)
comp (company-name, city)
manages (officer-name, manager-name)

Officer

officer-name,street,city
Rahul Singh,MG Road,Mumbai
Priya Mehta,Park Street,Bengaluru
Arjun Rao,MG Road,Mumbai
Kavita Shah,Civil Lines,Delhi
Rohan Patel,MG Road,Mumbai
Ananya Iyer,Brigade Road,Bengaluru
Manish Gupta,Residency Road,Pune
Sneha Nair,MG Road,Mumbai
Vikram Das,Sector 5,Kolkata
Divya Kumar,Jubilee Hills,Hyderabad

Placed

officer-name,company-name,salary
Rahul Singh,TCS,50000
Priya Mehta,TCS,60000
Arjun Rao,TCS,12000
Kavita Shah,Infosys,20000
Rohan Patel,Infosys,15000
Ananya Iyer,Infosys,8000
Manish Gupta,Amazon,40000
Sneha Nair,Amazon,45000
Vikram Das,Google,70000
Divya Kumar,Microsoft,30000

Comp

company-name,city
TCS,Mumbai
TCS,Bengaluru
Infosys,Mumbai
Infosys,Bengaluru
Infosys,Hyderabad
Amazon,Pune
Google,Kolkata
Microsoft,Hyderabad

Manages

officer-name,manager-name
Rahul Singh,Arjun Rao
Priya Mehta,Ananya Iyer
Arjun Rao,Rohan Patel
Kavita Shah,Manish Gupta
Rohan Patel,Rahul Singh
Ananya Iyer,Priya Mehta
Manish Gupta,Sneha Nair
Sneha Nair,Rahul Singh
Vikram Das,Divya Kumar
Divya Kumar,Vikram Das

1. Find the names of all officers who work for TCS.
2. Find the names and cities of residence of all officers who work for TCS.
3. Find the names, street addresses, and cities of all officers who work for TCS and earn more than Rs.10,000.
4. Find all officers who live in the same cities as the companies for which they work.
5. Find all officers who live in the same cities and on the same streets as their managers.
6. Find all officers who do not work for TCS.
7. Find all officers who earn more than each officer of Infosys.
8. Find all companies located in every city in which Infosys is located.
9. Find all officers who earn more than the average salary of all officers of their company.
10. Find the company that has the most officers.
11. Find the company that has the smallest payroll.
12. Find those companies whose officers earn a higher average salary than the average salary at TCS.

Problem Statement : 02
SQL Joins

Consider the given database schema:
Stud (studentid , studentname,instructorid,studentcity)
Inst(instructorid,Instructorname,instructorcity,specialization) Use all types of Joins

1. Find the instructor of each student.
2. Find the student who is not having any instructor.
3. Find the student who is not having any instructor as well as instructor who is not having student.
4. Find the students whose instructor's specialization is computer. 5.
5. Create a view containing total number of students whose instructor belongs to "Pune".

1,Rahul,101,Mumbai 2,Ananya,102,Bengaluru 3,Arjun,101,Mumbai 4,Kavita,103,Delhi 5,Rohan,104,Mumbai 6,Divya,102,Bengaluru 7,Manish,105,Pune 8,Sneha,101,Mumbai 9,Vikram,,Kolkata 10,Priya,106,Delhi	101,Dr. Sharma,Mumbai,Mathematics 102,Dr. Iyer,Bengaluru,Physics 103,Dr. Gupta,Delhi,Chemistry 104,Dr. Patel,Mumbai,Computer 105,Dr. Rao,Pune,Computer 106,Dr. Das,Kolkata,English 107,Dr. Mehta,Bengaluru,History 108,Dr. Nair,Delhi,Economics

Problem Statement : 03

SQL Joins

Consider the given database schema:
std (studentid , studentname,instructorid,studentcity)
Ist(instructorid,Instructorname,instructorcity,specialization) Use all types of Joins

11,Neha,201,Mumbai 12,Siddharth,202,Bengaluru 13,Meera,201,Pune 14,Akash,203,Delhi 15,Tanya,204,Chennai 16,Rahul,205,Mumbai 17,Simran,,Bengaluru 18,Karan,206,Kolkata 19,Ankit,207,Hyderabad 20,Prisha,204,Chennai	201,Dr. Kapoor,Mumbai,Mathematics 202,Dr. Menon,Bengaluru,Physics 203,Dr. Sharma,Delhi,Chemistry 204,Dr. Verma,Chennai,Computer 205,Dr. Das,Mumbai,Computer 206,Dr. Iyer,Kolkata,English 207,Dr. Gupta,Hyderabad,Biology 208,Dr. Nair,Pune,History

1. Find the instructor of each student.
2. Find the student who is not having any instructor.
3. Find the student who is not having any instructor as well as instructor who is not having student.
4. Find the students whose instructor's specialization is computer. 5.
5. Create a view containing total number of students whose instructor belongs to "Pune".

Problem Statement : 04

Problem:

A university wants to manage information about its students, courses, instructors, and departments. Each student can enroll in multiple courses, and each course can have many students. Each course is taught by an

instructor. Each instructor belongs to a single department, but a department can have multiple instructors. For each enrollment, the university wants to record the grade obtained by the student.

Tasks:

Identify entities, attributes, and keys.
Identify relationships (Enrolls, Teaches, BelongsTo) and their cardinalities.
Draw the ER diagram.
Convert the ER diagram into relational tables.

Problem Statement : 05

Hospital Management System Problem Statement

Problem:
A hospital wants to track patients, doctors, wards, and treatments. Each patient is admitted to a ward, and a ward can have multiple patients. Doctors can be assigned to multiple patients, and a patient can have multiple doctors. Each patient can undergo multiple treatments, and the hospital wants to store the treatment date for each treatment.

Tasks:

Identify entities, attributes, and keys.
Identify relationships (AdmittedTo, Assigned, Undergoes) and their cardinalities.
Draw the ER diagram.
Convert the ER diagram into relational tables.
Normalize the relational tables up to 3NF.

Problem Statement : 06

Online Shopping System Problem Statement

Problem:
An online shopping platform wants to manage customers, products, orders, and product categories. Each customer can place multiple orders. Each order can contain multiple products, and each product can appear in multiple orders. Each product belongs to a category. For each product in an order, the system wants to store the quantity and price at the time of order.

Tasks:

Identify entities, attributes, and keys.

Identify relationships (Places, Contains, BelongsTo) and their cardinalities.
Draw the ER diagram.
Convert the ER diagram into relational tables

Problem Statement : 0 7
PL/SQL block
Create a database with following schemas
Borrow(Roll, Name, Issue_Date, Book_Name Status)
Lib_Fine(Roll_no, Curr_Date, Amt)

Values:
Borrow(Roll, Name, Issue_Date, Book_Name, Status)
101, Rahul, 2025-10-15, "C Programming", 'I'
102, Ananya, 2025-10-01, "Data Structures", 'I'
103, Arjun, 2025-09-20, "DBMS Concepts", 'I'
104, Kavita, 2025-10-10, "Operating Systems", 'I'
105, Priya, 2025-09-25, "Python Programming", 'I'

1. Write a PL/SQL block to accept input for Borrower table.
2. Write a PL/SQL block using control structures to calculate fine by using the following rules:
a. check the number of days (from date of issue), if days are between 15 to 30 then fine amount will be Rs 5 per day
b. If no. of days>30, per day fine will be Rs 50 per day
c. for days less than 30, Rs. 5 per day.
After submitting the book, status will change from I to R. If condition of fine is true, then details will be stored into fine table.

Problem Statement : 08
PL/SQL block
Create a database with following schemas
Issue_Book(Roll, Name, Issue_Date, Book_Name Status)
Library_Fine(Roll_no, Curr_Date, Amt)

Values:
Borrow(Roll, Name, Issue_Date, Book_Name, Status)
106, Neha, 2025-09-10, "Java Programming", 'I'
107, Siddharth, 2025-10-05, "Algorithms", 'I'
108, Meera, 2025-09-15, "Computer Networks", 'I'
109, Akash, 2025-10-12, "Software Engineering", 'I'
110, Tanya, 2025-09-20, "Database Systems", 'I'

1. Write a PL/SQL block to accept input for Issue_Book table.
2. Write a PL/SQL block using control structures to calculate fine by using the following rules:
a. check the number of days (from date of issue), if days are between 15 to 30 then fine amount will be Rs 5 per day
b. If no. of days>30, per day fine will be Rs 50 per day
c. for days less than 30, Rs. 5 per day.
After submitting the book, status will change from I to R. If condition of fine is true, then details will be stored into fine table.

Problem Statement : 09

Database Trigger
Create a Library database with the schema
Pustak (Pustak_No, Title, Author, Publisher, Count).
Values:
Pustak_No, Title, Author, Publisher, Count
101, "C Programming", "Dennis Ritchie", "PHI", 10

102, "Data Structures", "Seymour Lipschutz", "McGraw Hill", 8
103, "Database Systems", "Korth", "TMH", 5
104, "Operating Systems", "Silberschatz", "Wiley", 7
105, "Python Programming", "Mark Lutz", "O'Reilly", 12

a. Create a table Lib_Audit with same fields as of Books and Date and status column
b. Create a before trigger to insert records into Lib_Audit table if there is deletion in Books table, insert date of deletion and status as deleted
C. Create a after trigger to insert records into Liby_Audit table if there is updation in Books table , insert date of updation and status as updated

Problem Statement : 10

Database Trigger
Create a Library database with the schema
Kitab (Kitab _No, Title, Author, Publisher, Count).
Values:
Kitab_*No, Title, Author, Publisher, Count*
106, "Java Programming", "James Gosling", "Oracle Press", 9
107, "Algorithms", "Robert Sedgewick", "Addison-Wesley", 6
108, "Computer Networks", "Andrew Tanenbaum", "Pearson", 7
109, "Software Engineering", "Ian Sommerville", "Pearson", 5
110, "Artificial Intelligence", "Stuart Russell", "Pearson", 4
a. Create a table Lib_Audit with same fields as of Books and Date and status column
b. Create a before trigger to insert records into Lib_Audit table if there is deletion in Books table, insert date of deletion and status as deleted
C. Create a after trigger to insert records into Liby_Audit table if there is updation in Books table , insert date of updation and status as updated

Problem Statement : 11
Create a table Books with fields: Book_ID, Title, Author, Publisher, Count.
Write a menu-driven or GUI program in Python/Java/PHP to connect to MySQL/Oracle/MongoDB and perform the following operations on the Books table:

Add a new book
Delete a book
Edit book details
View all books
Search a book by Book_ID

Problem Statement : 12

Create a table Students with fields: Student_ID, Name, City, Age, Department.
Write a menu-driven program in Python/Java/PHP to connect to MySQL/Oracle and perform the following operations on the Students table:

Add a new student
Delete a student
Edit student details
View all students
Search a student by Student_ID

Problem Statement : 13 MongoDB

Create a collection named library_books.
Insert 5 documents with fields: TITLE, DESCRIPTION, AUTHOR, LINK, CATEGORIES, and LIKES.
// 5 initial documents
{ "TITLE": "Python Basics", "DESCRIPTION": "Introduction to Python", "AUTHOR": "Ananya", "LINK": "http://example.com/python1", "CATEGORIES": ["Programming", "Python"], "LIKES": 120 }
{ "TITLE": "Advanced Python", "DESCRIPTION": "Deep dive into Python", "AUTHOR": "Raj", "LINK": "http://example.com/python2", "CATEGORIES": ["Programming", "Python"], "LIKES": 80 }
{ "TITLE": "Java Fundamentals", "DESCRIPTION": "Basics of Java", "AUTHOR": "Sneha", "LINK": "http://example.com/java1", "CATEGORIES": ["Programming", "Java"], "LIKES": 50 }
{ "TITLE": "Web Development", "DESCRIPTION": "HTML, CSS, JS", "AUTHOR": "Ananya", "LINK": "http://example.com/web", "CATEGORIES": ["Web", "HTML"], "LIKES": 30 }
{ "TITLE": "Data Science Intro", "DESCRIPTION": "Getting started with Data Science", "AUTHOR": "Raj", "LINK": "http://example.com/ds", "CATEGORIES": ["Data Science"], "LIKES": 200 }
Insert 1 more document with an additional field: USERNAME and REVIEWS.

// 1 additional document with extra fields
{ "TITLE": "Machine Learning", "DESCRIPTION": "ML Concepts", "AUTHOR": "Sneha", "LINK": "http://example.com/ml", "CATEGORIES": ["ML", "AI"], "LIKES": 150, "USERNAME": "user123", "REVIEWS": ["Excellent", "Very helpful"] }

Display all documents whose TITLE is 'Python Basics'.
Display all documents written by 'Ananya' or whose TITLE is 'Python Basics'.
Display all documents whose TITLE is 'Python Basics' and written by 'Ananya'.
Display all documents whose LIKES are greater than 15.
Display all documents whose LIKES are greater than 100 and whose TITLE is either 'Python Basics' or written by 'Ananya'.
Update the TITLE of 'Python Basics' document to 'Python Basics Overview'.
Delete the document whose TITLE is 'JavaScriptIntro'.
Display exactly two documents written by 'Ananya'.
Display the second document published by 'Ananya'.

Problem Statement : 14 MongoDB
Create a collection named **library_books**.
Insert the 5 documents above with fields `TITLE`, `DESCRIPTION`, `AUTHOR`, `LINK`, `CATEGORIES`, `LIKES`.
/ 5 initial documents
{ "TITLE": "JavaScript Essentials", "DESCRIPTION": "Learn JS basics", "AUTHOR": "Meera", "LINK": "http://example.com/js1", "CATEGORIES": ["Programming", "JavaScript"], "LIKES": 90 }
{ "TITLE": "NodeJS Guide", "DESCRIPTION": "Server-side JS", "AUTHOR": "Ananya", "LINK": "http://example.com/node", "CATEGORIES": ["Programming", "NodeJS"], "LIKES": 40 }
{ "TITLE": "ReactJS Intro", "DESCRIPTION": "Frontend framework", "AUTHOR": "Raj", "LINK": "http://example.com/react", "CATEGORIES": ["Web", "ReactJS"], "LIKES": 110 }
{ "TITLE": "Python for Data Analysis", "DESCRIPTION": "Data analysis using Python", "AUTHOR": "Sneha", "LINK": "http://example.com/python_data", "CATEGORIES": ["Python", "Data"], "LIKES": 75 }
{ "TITLE": "AI Basics", "DESCRIPTION": "Introduction to AI", "AUTHOR": "Meera", "LINK": "http://example.com/ai", "CATEGORIES": ["AI", "ML"], "LIKES": 60 }

Insert the additional document with extra fields USERNAME and REVIEWS.
{ "TITLE": "Deep Learning", "DESCRIPTION": "Neural networks", "AUTHOR": "Raj", "LINK": "http://example.com/dl", "CATEGORIES": ["AI", "DeepLearning"], "LIKES": 130, "USERNAME": "user456", "REVIEWS": ["Very informative", "Excellent guide"] }
Display all documents whose TITLE is 'ReactJS Intro'.
Display all documents written by 'Ananya' or whose TITLE is 'ReactJS Intro'.
Display all documents whose TITLE is 'ReactJS Intro' and written by 'Ananya'.
Display all documents whose LIKES are greater than 100.
Display all documents whose LIKES are greater than 100 and whose TITLE is either 'ReactJS Intro' or written by 'Raj'.
Update the TITLE of 'ReactJS Intro' document to 'ReactJS Overview'.
Delete the document whose TITLE is 'AI Basics'.
Display exactly two documents written by 'Meera'.
Display the second document published by 'Meera'.

Problem Statement : 15 MongoDB
Create a collection named tech_articles. Insert the 5 documents above with fields ArticleTitle, ArticleSummary, Writer, ArticleURL, Topics, LikesCount.
Values:
{ "BookTitle": "Python Basics", "BookDescription": "Introduction to Python", "AuthorName": "Ananya", "BookLink": "http://example.com/python1", "Categories": ["Programming","Python"], "BookLikes": 120 }
{ "BookTitle": "Advanced Python", "BookDescription": "Deep dive into Python", "AuthorName": "Raj", "BookLink": "http://example.com/python2", "Categories": ["Programming","Python"], "BookLikes": 80 }
{ "BookTitle": "Java Fundamentals", "BookDescription": "Basics of Java", "AuthorName": "Sneha", "BookLink": "http://example.com/java1", "Categories": ["Programming","Java"], "BookLikes": 50 }
{ "BookTitle": "Web Development", "BookDescription": "HTML, CSS, JS", "AuthorName": "Ananya", "BookLink": "http://example.com/web", "Categories": ["Web","HTML"], "BookLikes": 30 }
{ "BookTitle": "Data Science Intro", "BookDescription": "Getting started with Data Science", "AuthorName": "Raj", "BookLink": "http://example.com/ds", "Categories": ["Data Science"], "BookLikes": 200 }
Find the number of books published by 'Ananya'.
Find the books which have minimum likes and maximum likes published by 'Ananya'.
Find the average number of likes of the books published by 'Ananya'.
Find the first and last book published by 'Ananya'.
Create an index on AuthorName.
Display the books published by 'Ananya' and check if the query uses the index we have created.